Day 079 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Represent Ratios with Tape Diagrams

Standards: 6.RP.A.3a

Learning target: I can use equal-size tape sections to solve a ratio problem.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. For a ratio a:b, draw a sections for one quantity and b sections for the other. A total covers a+b sections. Divide the total by a+b to find one section, then multiply by each part count.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

Day 079 | Unit 06 | Lesson

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01 / Learn

For a ratio a:b, draw a sections for one quantity and b sections for the other. A total covers a+b sections. Divide the total by a+b to find one section, then multiply by each part count.

Worked example 1

The ratio of oak trees to pine trees is 6:5. There are 66 trees altogether. Draw a tape diagram and find each count.

1. For a ratio a:b, draw a sections for one quantity and b sections for the other.

2. A total covers a+b sections.

3. 11 equal sections total

4. each section is 66 ÷ 11 = 6.

5. State the result in the form requested: 36 oaks and 30 pines.

Answer: 36 oaks and 30 pines

Worked example 2

The ratio of paper flags to fabric flags is 3:2. There are 20 flags. Part A: draw a tape diagram and enter each count. Part B: a student says there are 13 paper flags. Use your model to explain the error.

1. For a ratio a:b, draw a sections for one quantity and b sections for the other.

2. There are 5 equal sections.

3. Each is 20 ÷ 5 = 4.

4. Multiply the section counts by 4.

Answer: 12 paper; 8 fabric; the claim is 1 paper flag too many.

Day 079 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

The ratio of oak trees to pine trees is 2:6. There are 16 trees altogether. Draw a tape diagram and find each count.

02 / Constructed response

The ratio of oak trees to pine trees is 6:3. There are 54 trees altogether. Draw a tape diagram and find each count.

03 / Constructed response

The ratio of oak trees to pine trees is 2:5. There are 35 trees altogether. Draw a tape diagram and find each count.

04 / Constructed response

The ratio of oak trees to pine trees is 8:6. There are 28 trees altogether. Draw a tape diagram and find each count.

05 / Constructed response

The ratio of oak trees to pine trees is 5:3. There are 48 trees altogether. Draw a tape diagram and find each count.

06 / Constructed response

The ratio of oak trees to pine trees is 5:7. There are 24 trees altogether. Draw a tape diagram and find each count.

07 / Constructed response

The ratio of oak trees to pine trees is 6:7. There are 52 trees altogether. Draw a tape diagram and find each count and the difference between the counts.

08 / Model and error analysis

The ratio of paper flags to fabric flags is 4:4. There are 16 flags. Part A: draw a tape diagram and enter each count. Part B: a student says there are 9 paper flags. Use your model to explain the error.

09 / Model and error analysis

The ratio of paper flags to fabric flags is 7:5. There are 36 flags. Part A: draw a tape diagram and enter each count. Part B: a student says there are 22 paper flags. Use your model to explain the error.

10 / Constructed response

The ratio of oak trees to pine trees is 5:3. There are 32 trees altogether. Draw a tape diagram and find each count and the difference between the counts.

Day 079 | Unit 06 | Answer key

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1. 4 oaks and 12 pines

8 equal sections total; each section is 16 ÷ 8 = 2.

6.RP.A.3a | Constructed response

2. 36 oaks and 18 pines

9 equal sections total; each section is 54 ÷ 9 = 6.

6.RP.A.3a | Constructed response

3. 10 oaks and 25 pines

7 equal sections total; each section is 35 ÷ 7 = 5.

6.RP.A.3a | Constructed response

4. 16 oaks and 12 pines

14 equal sections total; each section is 28 ÷ 14 = 2.

6.RP.A.3a | Constructed response

5. 30 oaks and 18 pines

8 equal sections total; each section is 48 ÷ 8 = 6.

6.RP.A.3a | Constructed response

6. 10 oaks and 14 pines

12 equal sections total; each section is 24 ÷ 12 = 2.

6.RP.A.3a | Constructed response

7. 24 oaks and 28 pines; difference 4

13 equal sections total; each section is 52 ÷ 13 = 4.

6.RP.A.3a | Constructed response

8. 8 paper; 8 fabric; the claim is 1 paper flag too many.

There are 8 equal sections. Each is 16 ÷ 8 = 2. Multiply the section counts by 2.

6.RP.A.3a | Model and error analysis

9. 21 paper; 15 fabric; the claim is 1 paper flag too many.

There are 12 equal sections. Each is 36 ÷ 12 = 3. Multiply the section counts by 3.

6.RP.A.3a | Model and error analysis

10. 20 oaks and 12 pines; difference 8

8 equal sections total; each section is 32 ÷ 8 = 4.

6.RP.A.3a | Constructed response